

Symplectic leaves of Coulomb branches

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Fix a reductive group G and a representation $\mathbf{N}$. The **Coulomb branch** $\mathcal{M}(G, \mathbf{N})$ is a symplectic variety constructed via quantum field theory.

Two main questions

1. How can we describe Coulomb branches geometrically in a way that is *uniform* across the rational, K-theoretic, and elliptic settings?
2. $\mathcal{M}(G, \mathbf{N})$ has symplectic singularities, so it has finitely many **symplectic leaves**. What is their classification, and what are their transverse slices?

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An important 3d theory: given G a connected reductive group and $\mathbf{E}$ a symplectic representation of G , we have a theory $\mathcal{T}(G, \mathbf{N})$ given by gauging the 3d σ -model with target $T^*\mathbf{N}$ for a representation $\mathbf{N}$.

We have two natural topological twists $\mathcal{Q}_A, \mathcal{Q}_B$. We call theories **3-d mirror dual** if they are equivalent in a way that swaps $\mathcal{Q}_A \leftrightarrow \mathcal{Q}_B$.

Since theories are 3d, their local operators are commutative rings equipped with Poisson brackets. We think of these as functions on algebraic varieties called the **Coulomb** and **Higgs** branches $\mathfrak{M}_C$ and $\mathfrak{M}_H$.

The moduli of vacua $\mathfrak{M}_H(G, \mathbf{E}) = \text{Spec } Z_B(S^2)$ for the B -twist (the **Higgs branch**) was studied by Hitchin, Karlhede, Lindström and Roček and interpreted as a hyperkähler quotient $\mathbf{E} // G$.

The moduli of vacua for the A-twist (the **Coulomb branch**) proved much more mysterious. Denote this $\mathfrak{M}_C(G, \mathbf{N}) = \text{Spec } Z_A(S^2)$ (I'm assuming $\mathbf{E} = T^*\mathbf{N}$).

- ▶ $\mathbb{C}[\mathfrak{M}_C(G, \mathbf{N})]$ is a subalgebra of the fraction field $\mathbb{C}(T^*T^\vee)^W$,
- ▶ obtained by multiplying e^λ , a character on $T^\vee$, by certain rational functions (roughly with denominator depending on G and numerator depending on $\mathbf{N}$).

Taylor series $\mathbf{C} = \mathbb{C}[[t]]$	$\mathbf{G} = G[[t]]$	$\mathbf{N} = \mathbf{N}[[t]]$
Laurent series $\mathcal{C} = \mathbb{C}((t))$	$\mathcal{G} = G((t))$	$\mathcal{N} = \mathbf{N}((t))$

Relevant spaces:

$$\mathbf{Y} = \mathbf{N}/\mathbf{G} \quad \mathcal{Y} = \mathcal{N}/\mathcal{G}$$

These can be interpreted as spaces of principal G bundles with a section of the associated $\mathbf{N}$ -bundle on D and D^* .

Thus, the fiber product $\mathbf{Y} \times_{\mathcal{Y}} \mathbf{Y}$ is the space of such bundles on the “raviolo” gluing two copies of D along D^* .

Definition (BFN)

The **Coulomb branch** is the spectrum $\mathcal{M} = \mathrm{Spec} H_*^{BM}(\mathbf{Y} \times_{\mathcal{Y}} \mathbf{Y})$.

Most important case: let Γ be a quiver, and $\mathbf{v}: I = \mathcal{V}(\Gamma) \rightarrow \mathbb{Z}_{\geq 0}$ a dimension vector. Let

$$N_{\mathbf{v}} = \bigoplus_{i \rightarrow j} \mathrm{Hom}(\mathbb{C}^{v_i}, \mathbb{C}^{v_j}) \quad G_{\mathbf{v}} = \prod_i GL(\mathbb{C}^{v_i})$$

These recover interesting varieties, always symplectic:

- ▶ my favorite: nilcone of $\mathfrak{gl}_n$
- ▶ more generally, Slodowy slices in type A
- ▶ symmetric power $\mathrm{Sym}^n(\mathbb{C}^2/\mathbb{Z}_{\ell})$

The BFN construction uses Borel–Moore homology. The same construction works uniformly for three choices of a 1-dimensional algebraic group B :

B	Cohomology theory	Base $\mathcal{B}_G$
$\mathbb{G}_a = \operatorname{Spec} \mathbb{C}[x]$	Rational (BFN)	$\mathfrak{t}/W$
$\mathbb{G}_m = \operatorname{Spec} \mathbb{C}[z^{\pm 1}]$	K-theoretic	T/W
E (elliptic curve)	Elliptic	$E \otimes_{\mathbb{Z}} X_*(T)/W$

In each case, $\mathcal{M}(G, \mathbf{N})$ is a variety over the **base** $\mathcal{B}_G$. For each weight φ of $\mathbf{N}$ (or root α of G), there is an **Euler class** $\operatorname{Eu}(\varphi)$ on $\mathcal{B}_T$ defined by pulling back the divisor $\{\operatorname{id}\}$ by the induced map $\varphi: \mathcal{B}_T \rightarrow \mathcal{B}_{\mathbb{G}_m}$ for a weight.

For example: $G = \mathbb{G}_m$. $\mathcal{M}$ is birational to $B \times \mathbb{G}_m$. The weights of $\mathbf{N}$ are $n_1, \dots, n_k \in \mathbb{Z}$. Let

$$\mathrm{Eu}^\pm = \prod_{\pm n_i > 0} \mathrm{Eu}(n_i)^{|n_i|}.$$

In $B \times \mathbb{P}_z^1$, consider the subschemes $X^\pm = Z(\mathrm{Eu}^\pm, z^{\mp 1})$.

Theorem

$\mathcal{M}(\mathbb{G}_m, \mathbf{N})$ is the affinization of the blowup of $B \times \mathbb{P}_z^1$ at $X^+ \cup X^-$ with strict transforms of $B \times \{0\}$ and $B \times \{\infty\}$ removed.

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General case: need to do a similar blowup for each of the lines $\ell_1, \dots, \ell_k$ spanned by weights of $\mathbf{N}$.

Teleman gluing

If the weights of $\mathbf{N}$ are generic enough, then all the new divisors can be written as the images of fibers over points in $\mathcal{B}_T$ under a birational map, so $\mathcal{M}(T, \mathbf{N})$ is the affinization of $\mathcal{B}_T \times T^\vee$ glued to itself along this birational map.

Unifies rational, K-theoretic, and elliptic cases. Local-to-global: K-theoretic and elliptic Coulomb branches look locally like the rational one (base $\cong$ tangent space locally).

Passing from T to G requires a further birational modification: if no root lies in ℓ_i , then we blow up the subscheme in $\mathcal{B}_T \times T^\vee$ given by $Z(\text{Eu}(\alpha), e^{\alpha^\vee} - 1)$ for each root α , and remove the strict transform of $Z(\text{Eu}(\alpha))$.

Theorem

The Coulomb branch $\mathcal{M}(G, \mathbf{N})$ is the affinization of this birational modification of $\mathcal{B}_T \times T^\vee$ modulo W .

If roots do lie in ℓ_i , then fixable, but need to be more careful.

Put differently: $\mathcal{M}(G, \mathbf{N})$ is a normal variety, so we can describe it by giving Weil divisors. These are only different from $(\mathcal{B}_T \times T^\vee)/W$ over the divisors where Euler classes of roots or weights vanish, and the modification is obtained from those divisors in a uniform geometric way.

- ▶ Over a weight divisor where no root vanishes, we have the two divisors coming from the blowups as in the abelian case.
- ▶ Over a root divisor, we usually only have one divisor, since the reflection combines them (though there are a couple of exceptions where there are two).

Important consequence: removing a divisor has the same effect as removing the corresponding weight from $\mathbf{N}$ or root from G .

Given $\lambda \in \mathfrak{t}$, there is a commuting subgroup $G_\lambda \subset G$ with Weyl group W_λ and fixed representation $\mathbf{N}^\lambda \subset \mathbf{N}$.

Let $V \subset \mathcal{B}_T$ be the complement of the divisors corresponding to roots or weights which don't vanish at $W\lambda$.

Theorem

The varieties $\mathcal{M}(G, \mathbf{N})$ and $\mathcal{M}(G_\lambda, \mathbf{N}^\lambda)$ over U/W_λ are isomorphic.

$$\begin{array}{ccccc}
 \mathcal{M}(G, \mathbf{N}) & \longleftarrow & \mathcal{M}^\circ(G_\lambda, \mathbf{N}_\lambda) & \hookrightarrow & \mathcal{M}(G_\lambda, \mathbf{N}_\lambda) \\
 \downarrow & & \downarrow & & \downarrow \\
 \mathcal{B}_G = \mathcal{B}_T/W & \longleftarrow & V/W_\lambda & \hookrightarrow & \mathcal{B}_{G_\lambda} = \mathcal{B}_T/W_\lambda
 \end{array}$$

Bielawski–Foscolo proposed describing Coulomb branches via **transverse W -Hilbert schemes**.

Closure of free W -orbits in the moduli of subschemes of $\mathcal{M}(T, \mathbf{N})$ that push forward to subschemes of $\mathcal{B}_T$.

Theorem

The functions on $\mathrm{Hilb}_\pi^W(X)$ are the subring of $\mathbb{k}(X)^W$ generated by $(\frac{1}{\Delta}\mathbb{k}[X])^W$.

For $G = PGL(2)$ or a quiver gauge theory: exactly $\mathcal{M}(G, \mathbf{N})$.

The issue for general G : there “aren’t enough” sign-invariant functions on $\mathcal{M}(T, \mathbf{N})$ to generate the functions on $\mathcal{M}(G, \mathbf{N})$.

Corrected theorem

Assume that G has no simple factor isomorphic to $Sp(2n)$.

Replace $\mathcal{M}(T, \mathbf{N})$ by the open subset $\mathcal{M}^\bullet(T, \mathbf{N})$ obtained by removing the locus over intersections of bad divisors. Then:

$$\mathcal{M}^\bullet(G, \mathbf{N}) \cong \mathrm{Hilb}_\pi^W(\mathcal{M}^\bullet(T, \mathbf{N}))$$

and $\mathcal{M}(G, \mathbf{N})$ is the affinization of this variety.

A **symplectic singularity** has a canonical stratification into finitely many **symplectic leaves**: locally closed smooth symplectic subvarieties. These are the orbits of the Hamiltonian vector fields.

Fundamental questions for $\mathcal{M}(G, \mathbf{N})$:

1. Index the leaves (how many are there? how to describe them?).
2. Identify the **transverse slice** to each leaf (another symplectic singularity).

In the abelian case $G = T$, the answer is well-known from the Higgs branch description:

- ▶ Leaves are in bijection with **co-loop-free flats** of $\mathfrak{t}$.
 - ▶ A **flat** F is an intersection of hyperplanes defined by weights of $\mathbf{N}$; this is precisely the stabilizer of a point in $\mathbf{N}$.
 - ▶ A **co-loop** of F is a weight φ such that there exists a cocharacter λ of T with $\lambda(\varphi) = 1$ and $\lambda(\psi) = 0$ for all other weights ψ vanishing on F .
- ▶ If λ is generic in this stabilizer T_0 , then the slice to this leaf is $\mathcal{M}(T/T_0, \mathbf{N}^\lambda)$.

If a flat has a co-loop, it still gives a subspace, but not a symplectic leaf: consider $\mathcal{M}(\mathbb{G}_m, \mathbb{C}) \cong \mathbb{C}^2$.

Quite suggestive, but much simpler than the general case.

The distinction between good, bad and ugly theories will be important for us:

- ▶ **good**: the scaling action contracts to a unique fixed point, which is a 0-dimensional symplectic leaf. Example: $\mathcal{M}(\mathbb{G}_m, \mathbb{C}^2) \cong \mathbb{C}^2/\mathbb{Z}_2$, abelian theories with finite kernel, no co-loops.
- ▶ **ugly**: the scaling action contracts to a fixed point, but this fixed point lies in a positive-dimensional leaf (in fact a symplectic vector space). Example: $\mathcal{M}(\mathbb{G}_m, \mathbb{C}) \cong \mathbb{C}^2$, abelian theories with finite kernel and co-loops.
- ▶ **bad**: the scaling action doesn't contract to a single fixed point. Example: $\mathcal{M}(\mathbb{G}_m, 0) \cong \mathbb{C} \times \mathbb{C}^\times$, abelian theories with infinite kernel.

Abelian case: a leaf appears exactly when $(T/T_0, \mathbf{N}^\lambda)$ is good.

There is a natural **integrable system**:

$$\varrho: \mathcal{M}(G, \mathbf{N}) \longrightarrow \mathcal{B}_G,$$

induced by the map from the pure Coulomb branch $\mathcal{M}(G)$, which is a group scheme over $\mathcal{B}_G$.

Over the open dense set of $\mathcal{B}_G$ where no weights or roots vanish, the fiber of ϱ is a torsor for the dual torus $T^\vee$.

Can break the classification of leaves into two steps:

1. Classify the possible images of leaves in $\mathcal{B}_G$.
2. For each image, classify the leaves lying over it.

Definition

A **flat** of the generalized weight hyperplane arrangement in $\mathfrak{t}$ is an intersection of a subset of the hyperplanes H_φ defined by weights of $\mathbf{N}$; these are exactly the stabilizers of points in $\mathbf{N}$.

For a flat $U \subset \mathfrak{t}$ with generic point λ :

- ▶ $G_\lambda = C_G(\lambda)$ is a **Levi subgroup** of G
- ▶ $\mathbf{N}^\lambda = \mathbf{N}^\lambda$ is the fixed subspace of λ (a G_λ -representation)
- ▶ $G_\lambda^0 = \ker^0(G_\lambda \rightarrow GL(\mathbf{N}^\lambda))$; $G_\lambda^1 = G_\lambda / G_\lambda^0$ almost faithful on $\mathbf{N}^\lambda$
- ▶ $\mathfrak{t}_\lambda^0 = \text{Lie algebra of maximal torus of } G_\lambda^0$ (of dimension k)

Slice theorem (local structure)

Around the fiber of ϱ over λ , $\mathcal{M}(G, \mathbf{N})$ is étale locally symplectomorphic to $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda) \times \mathbb{A}^{2k}$.

We have to be a little more careful about global structure, though.

- ▶ There might be a Galois group $\bar{\Gamma}_\lambda$ that appears in the comparison of $\mathcal{M}(G, \mathbf{N})$ and $\mathcal{M}(G_\lambda, \mathbf{N}^\lambda)$ over U .
- ▶ There is an isogeny $G_\lambda \rightarrow G_\lambda^1 \times \bar{G}_\lambda^0$ for $\bar{G}_\lambda^0$ a finite quotient; let Z_λ be the kernel of this map.

Lemma

The product $\mathcal{M}(G_\lambda^1 \times \bar{G}_\lambda^0, \mathbf{N}^\lambda) \cong \mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda) \times \mathcal{M}(\bar{G}_\lambda^0, 0)$ admits a free action of $Z_\lambda^\vee$, and $\mathcal{M}(G_\lambda, \mathbf{N}^\lambda)$ is the quotient by this action.

The symplectic leaves of $\mathcal{M}(G_\lambda, \mathbf{N}^\lambda)$ are in bijection with $Z_\lambda^\vee$ -orbits on the symplectic leaves of $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$, and the transverse slice to a leaf in $\mathcal{M}(G_\lambda, \mathbf{N}^\lambda)$ is the same as the transverse slice to any leaf in the corresponding orbit in $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$.

Theorem (W.)

1. *The image of any symplectic leaf of $\mathcal{M}(G, \mathbf{N})$ under ϱ is the image in $\mathfrak{t}/W$ of a flat U .*
2. *The leaves with image U are in bijection with $Z_\lambda^\vee \rtimes \bar{\Gamma}_\lambda$ -orbits on 0-dimensional leaves of $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$.*
3. *If U is **good** (i.e., $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$ is good), there is a **unique** leaf with image U , and its **transverse slice** is isomorphic to $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$.*
4. *If $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$ has no 0-dimensional leaves, there is **no** leaf with image U . Important case: $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$ is ugly.*

Lemma

A leaf L with image U lies in the closure of L' with image U' if and only if:

- 1. $U \subset U'$, and*
- 2. the corresponding closure relation holds inside $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$.*

In particular, if both U and U' are good, then $L \subset \overline{L'}$ if and only if $U \subset U'$.

This gives a complete description of the **closure poset** on good leaves: inclusion on good flats.

Consider the case $G = \mathrm{GL}_n$, $\mathbf{N} = \mathfrak{gl}_n$.

1. $\mathcal{M}(G, \mathbf{N}) \cong \mathrm{Sym}^n(\mathbb{C} \times \mathbb{C}^\times)$.
2. Flats: indexed by partitions $\lambda \vdash n$ (groups of equal coordinates).
3. For $\lambda = (n_1, \dots, n_k)$:

$$G_\lambda^1 = \prod \mathrm{PGL}_{n_i}, \quad \mathbf{N}^\lambda = \bigoplus \mathfrak{gl}_{n_i}.$$

4. Residual theory $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$ has $\prod n_i$ zero-dimensional leaves, but $Z^\vee$ acts simply transitively $\Rightarrow$ **one leaf per partition**.

Now modify to $G = \mathrm{GL}_n$, $\mathbf{N} = \mathbb{C}^n \oplus \mathfrak{gl}_n$.

1. $\mathcal{M}(G, \mathbf{N}) \cong \mathrm{Sym}^n(\mathbb{C}^2)$.
2. Flats: indexed by an integer $n_0 \leq n$ (number of coordinates equal to 0) and by partitions $\lambda \vdash (n - n_0)$.
3. Slice described by

$$G_\lambda^1 = \mathrm{GL}_{n_0} \times \prod \mathrm{PGL}_{\lambda_i}, \quad \mathbf{N}^\lambda = \mathbb{C}^{n_0} \oplus \mathfrak{gl}_{n_0} \oplus \bigoplus \mathfrak{gl}_{\lambda_i}.$$

$(\mathrm{GL}_{n_0}, \mathbb{C}^{n_0} \oplus \mathfrak{gl}_{n_0})$ is **ugly** for all $n_0 > 0$, so $\mathcal{M}(G_\lambda^1, \mathbf{N}^\lambda)$ has no 0-dimensional leaves in this case.

4. For $n_0 = 0$, same flats as above give the same Hasse diagram.

One more: $G = GL_n$, $\mathbf{N} = (\mathbb{C}^n)^{\oplus \ell} \oplus \mathfrak{gl}_n$ ($\ell > 1$).

1. $\mathcal{M}(G, \mathbf{N}) \cong \text{Sym}^n(\mathbb{C}^2/\mathbb{Z}_\ell)$.
2. Flats indexed by pairs (n_0, λ) with $n_0 \leq n$, $\lambda \vdash (n - n_0)$.
3. Slice described by

$$G_\lambda^1 = GL_{n_0} \times \prod PGL_{\lambda_i}, \quad \mathbf{N}^\lambda = (\mathbb{C}^{n_0})^{\oplus \ell} \oplus \mathfrak{gl}_{n_0} \oplus \bigoplus \mathfrak{gl}_{\lambda_i}.$$

4. For $n_0 > 0$, $(GL_{n_0}, (\mathbb{C}^{n_0})^{\oplus \ell} \oplus \mathfrak{gl}_{n_0})$ is **good**. So there is a leaf for each pair (n_0, λ) , given by n_0 points at the origin and $n - n_0$ points in $\mathbb{C}^2/\mathbb{Z}_\ell$ with multiplicities given by λ .

These are special cases of quiver gauge theories.

$$G_{\mathbf{v}} = \left(\prod_i \mathrm{GL}_{v_i} \right) / \mathbb{C}^\times \quad \mathbf{N}_{\mathbf{v}} = \bigoplus_{i \rightarrow j} \mathrm{Hom}(\mathbb{C}^{v_i}, \mathbb{C}^{v_j})$$

A cocharacter for a quiver gauge theory is a grading of the corresponding quiver representation, so the flat is described by the dimension vectors of the graded pieces.

“Fission and decay”

Writing $\mathbf{v} = \sum \mathbf{v}^{(i)}$ as a sum of dimension vectors:

1. “Decay”: If $\mathbf{v}^{(i)}$ is a unit vector, trivial contribution to the slice.
2. “Fission”: If more than one of the summands $\mathbf{v}^{(i)}$ is not a unit vector, then the slice is the product of contributions from the nontrivial $\mathbf{v}^{(i)}$.

Examples

Note that the framing is expressed by adding a new vertex, so it all goes with one summand.

For a framed Dynkin quiver, the only contributions come from dimension vectors of dominant weights, with unit vectors “decayed off.”

This matches the results of Muthiah and Weekes.

For example: type A nilcone

- ▶ Theory is $G = \mathrm{GL}_1 \times \cdots \times \mathrm{GL}_{n-1}$, $\mathbf{N} = \bigoplus_{i=1}^{n-1} \mathrm{Hom}(\mathbb{C}^i, \mathbb{C}^{i+1})$.
- ▶ The good residual theories correspond to dimension vectors v_i with $v_1 \leq v_2 - v_1 \leq \cdots \leq v_{n-1} - v_{n-2} \leq n - v_{n-1}$. These give a partition $\lambda_i = v_{n-i+1} - v_{n-i}$.

The corresponding leaf has Jordan type λ ; the slice $\mathcal{M}(\mathrm{PGL}_{\mathbf{v}}, \mathbf{N}_{\mathbf{v}})$ is the Slodowy slice to the nilpotent orbit of type λ .

For an affine type A quiver, we can also “fission” off multiples of the imaginary root δ , contributing the centered Uhlenbeck space $\mathcal{M}(PGL_{k\delta}, \mathbf{N}_{k\delta})$ to the slice, matching Nakajima–Takayama.

Let $\text{Leaf}(\mathcal{M}_{\text{Coulomb}})$ and $\text{Leaf}(\mathcal{M}_{\text{Higgs}})$ denote the leaf sets of the Coulomb and Higgs branches.

Definition

Define $\mathbf{CH} \subset \text{Leaf}(\mathcal{M}_{\text{Coulomb}}) \times \text{Leaf}(\mathcal{M}_{\text{Higgs}})$ by $(L, L') \in \mathbf{CH}$ if, for a generic point $[x] \in L'$, the image of L under ϱ equals the flat $\mathfrak{t}_x$ (Lie algebra of the torus of the stabilizer G_x).

Physics expects an **order-reversing bijection** between leaves, but several obstructions arise:

1. Multiple Coulomb leaves over the same flat.
2. A flat may not arise as $\mathfrak{t}_x$ for any Higgs leaf.
3. Multiple Higgs leaves with the same stabilizer torus.

Example: $G = \text{GL}_n$, $\mathbf{N} = \mathbb{C}^n \oplus \mathfrak{gl}_n$. Every Higgs leaf has $\mathfrak{t}_x = \mathfrak{t}$, so only the open Coulomb leaf appears.

Conjecture

*There is a natural notion of **special** leaf on each side such that **CH** restricts to a bijection between the special leaves of $\mathcal{M}_{\text{Coulomb}}$ and $\mathcal{M}_{\text{Higgs}}$.*

*The special Coulomb leaves are exactly those whose image under ϱ is a **good flat**.*

Evidence:

- ▶ **Toric case** ($G = T$): all leaves are special; bijection is obvious.
- ▶ **Dynkin quivers**: all leaves are special; bijection is obvious.
- ▶ $G = \text{GL}_n$, $\mathbf{N} = (\mathbb{C}^n)^{\oplus \ell} \oplus \mathfrak{gl}_n$: special leaves are those with $\lambda = (1, \dots, 1)$. Order-reversing bijection with special Higgs leaves, but not all leaves.

1. **Zero-dimensional leaves in bad/ugly theories.** When does a bad flat admit a 0-d leaf? Often none exist, but sometimes they do. The example $G = PGL_n$, $\mathbf{N} = \mathfrak{pgl}_n$ shows that any number of zero-dimensional leaves is possible. Bourget *et. al.* conjecture that known affine Dynkin quiver examples are the only ones.
2. **Elliptic Coulomb branches.** No consensus definition exists yet. Our construction gives a candidate; comparison with BFN is pending more development in the elliptic literature.
3. **Noncotangent cases.** If $\mathbf{E}$ is a symplectic representation of G that is not a cotangent, then $\mathcal{M}(G, \mathbf{E})$ can still be defined if certain obstructions vanish. Leaf structure is unknown, but should be similar.